

SAHRDAYA COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
Module wise model questions

EET401: ADVANCED CONTROL SYSTEMS
MODULE 1

1. Explain any three advantages of state model as compared to transfer function model.

1. The state space analysis is applicable to any type of systems. They can be used for modelling and analysis of linear & non-linear systems, time invariant & time variant systems and multiple input & multiple output systems.
2. The state space analysis can be performed with initial conditions.
3. The variables used to represent the system can be any variables in the system.
4. Using this analysis the internal states of the system at any time instant can be predicted.

2.

What are the drawbacks in transfer function model analysis?

1. Transfer function is defined under zero initial conditions.
2. Transfer function is applicable to linear time invariant systems.
3. Transfer function analysis is restricted to single input and output systems.
4. Does not provides information regarding the internal state of the system.

3.

What is state and state variable?

The state is the condition of a system at any time instant, t . A set of variable which describes the state of the system at any time instant are called state variables.

4.

Write the state model of n^{th} order system?

The state model of a system consists of state equation and output equation. The state model of a n^{th} order system with m -inputs and p -outputs are

$$\dot{X}(t) = AX(t) + BU(t) \quad \dots \text{state equation.}$$

$$Y(t) = CX(t) + DU(t) \quad \dots \text{Output equation.}$$

where $X(t)$ = State vector of order $(n \times 1)$; $U(t)$ = Input vector of order $(m \times 1)$

A = System matrix of order $(n \times n)$; B = Input matrix of order $(n \times m)$

$Y(t)$ = Output vector of order $(p \times 1)$; C = Output matrix of order $(p \times n)$.

D = Transmission matrix of order $(p \times m)$.

5. The state model of a LTI system is given by $\dot{x} = Ax + Bu$ and $y = Cx + Du$. Obtain the expression for transfer function of the system

Solution : Given that $\dot{X}(t) = A X(t) + B U(t)$ (1)

and $Y(t) = C X(t) + D U(t)$ (2)

On taking laplace transform of equ(1) with zero initial conditions we get,

$$s X(s) = A X(s) + B U(s)$$

$$s X(s) - A X(s) = B U(s)$$

$$(sI - A) X(s) = B U(s)$$
(3)

On premultiplying equ(3) by $(sI - A)^{-1}$ we get

$$X(s) = (sI - A)^{-1} B U(s)$$
(4)

On taking laplace transform of equ(2) we get,

$$Y(s) = C X(s) + D U(s)$$
(5)

Substitute for $X(s)$ from equ(4) in equ(5) we get,

$$Y(s) = C [(sI - A)^{-1} B U(s)] + D U(s)$$

$$= [C (sI - A)^{-1} B U(s) + D] U(s)$$

$$\therefore \frac{Y(s)}{U(s)} = C (sI - A)^{-1} B U(s) + D$$
(6)

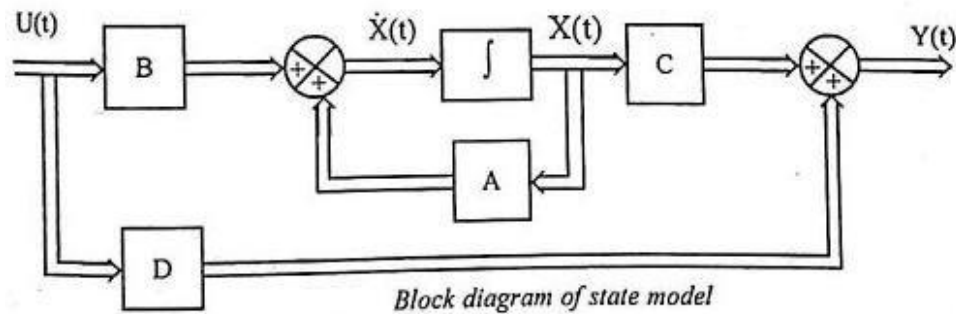
The equation (6) is the transfer function of the system.

6.

What is state diagram?

The pictorial representation of the state model of the system is called state diagram. The state diagram of the system can be either in block diagram or in signal flow graph form.

7. **Draw the block diagram representation of state model**



8. **Derive the state model of an armature controlled DC motor.**

9. Obtain the state model of a field controlled DC motor

10. **Obtain the state model of the system whose transfer function is given as**

$$\frac{Y(s)}{U(s)} = \frac{10}{s^3 + 4s^2 + 2s + 1}$$

SOLUTION

Method 1

$$\text{Given that, } \frac{Y(s)}{U(s)} = \frac{10}{s^3 + 4s^2 + 2s + 1} \quad \dots(4.1)$$

On cross multiplying the equ (4.10.1) we get,

$$Y(s)[s^3 + 4s^2 + 2s + 1] = 10 U(s) \quad \dots(4.1)$$

$$s^3 Y(s) + 4s^2 Y(s) + 2s Y(s) + Y(s) = 10 U(s)$$

On taking inverse laplace transform of equ (4.10.2) we get,(4.1)

$$\ddot{y} + 4\ddot{y} + 2\dot{y} + y = 10 u.$$

Let us define state variables as follows,

$$x_1 = y \quad ; \quad x_2 = \dot{y} \quad ; \quad x_3 = \ddot{y}$$

Put $\ddot{y} = \dot{x}_3$; $\dot{y} = x_3$; $\dot{y} = x_2$ and $y = x_1$ in the equation (4.10.3)

$$\therefore \dot{x}_3 + 4x_3 + 2x_2 + x_1 = 10u$$

$$\text{or } \dot{x}_3 = -x_1 - 2x_2 - 4x_3 + 10u.$$

The state equations are

$$\dot{x}_1 = x_2 \quad ; \quad \dot{x}_2 = x_3 \quad ; \quad \dot{x}_3 = -x_1 - 2x_2 - 4x_3 + 10u.$$

The output equation is $y = x_1$

The state model in the matrix form is,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} [u]$$

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

11.

Q4.11. Draw the signal flow graph representation of state model?

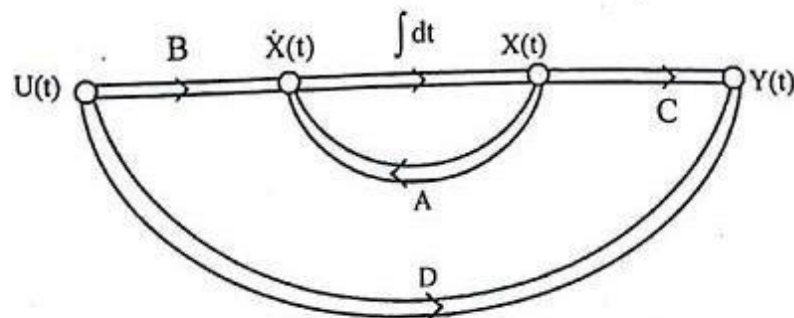


Fig Q4.11 : Signal flow graph of state model

Q4.19. A system is characterized by the differential equation,

$$\frac{d^2 y}{dt^2} + 10 \frac{dy}{dt} + 7y - u = 0$$

Determine its transfer function.

Solution

Given that, $\frac{d^2 y}{dt^2} + 10 \frac{dy}{dt} + 7y - u = 0$ (1)

On taking laplace transform of equ(1) with zero initial conditions we get,

$$s^2 Y(s) + 10 s Y(s) + 7 Y(s) - U(s) = 0$$

$$(s^2 + 10 s + 7) Y(s) = U(s)$$

$$\therefore \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 10s + 7} \quad \text{.....(2)}$$

The equ(2) is the transfer function of the system.

Q4.22. What are phase variables?

The phase variables are defined as those particular state variables which are obtained from one of the system variable and its derivatives. Usually the variable used is the system output and the remaining state variables are then derivatives of the output.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & -a_{n-3} & \dots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ b \end{bmatrix} [u]$$

$$y = [1 \quad 0 \quad 0 \quad \dots \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

Construct a state model for a system characterized by the differential equation,

$$\frac{d^3y}{dt^3} + 6 \frac{d^2y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0.$$

Give the block diagram representation of the state model.

Let us choose y and their derivatives as state variables. The system is governed by third order differential equation and so the number of state variables are three.

The state variables x_1 , x_2 and x_3 are related to phase variables as follows

$$x_1 = y$$

$$x_2 = \frac{dy}{dt} = \dot{x}_1$$

$$x_3 = \frac{d^2y}{dt^2} = \dot{x}_2$$

Put $y = x_1$, $\frac{dy}{dt} = x_2$ and $\frac{d^2y}{dt^2} = x_3$ and $\frac{d^3y}{dt^3} = \dot{x}_3$ in the given equation,

$$\therefore \dot{x}_3 + 6x_3 + 11x_2 + 6x_1 + u = 0$$

$$\text{or } \dot{x}_3 = -6x_1 - 11x_2 - 6x_3 - u.$$

The state equations are

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -6x_1 - 11x_2 - 6x_3 - u$$

On arranging the state equations in the matrix form we get,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} [u]$$

Here, y = output

But, $y = x_1$

Controllable Canonical Form. The following state-space representation is called a controllable canonical form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u \quad (9-3)$$

$$y = [b_n - a_n b_0 \quad b_{n-1} - a_{n-1} b_0 \quad \cdots \quad b_1 - a_1 b_0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u \quad (9-4)$$

Observable Canonical Form. The following state-space representation is called an observable canonical form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ \vdots \\ b_1 - a_1 b_0 \end{bmatrix} u \quad (9-5)$$

$$y = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u \quad (9-6)$$

Diagonal Canonical Form. Consider the transfer-function system defined by Equation (9-2). Here we consider the case where the denominator polynomial involves only distinct roots. For the distinct-roots case, Equation (9-2) can be written as

$$\begin{aligned} \frac{Y(s)}{U(s)} &= \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{(s + p_1)(s + p_2) \cdots (s + p_n)} \\ &= b_0 + \frac{c_1}{s + p_1} + \frac{c_2}{s + p_2} + \cdots + \frac{c_n}{s + p_n} \end{aligned} \quad (9-7)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & & & 0 \\ & -p_2 & & \\ & & \ddots & \\ 0 & & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \quad (9-8)$$

$$y = [c_1 \ c_2 \ \cdots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u \quad (9-9)$$

EXAMPLE 9-1 Consider the system given by

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2}$$

Obtain state-space representations in the controllable canonical form, observable canonical form, and diagonal canonical form.

Controllable Canonical Form:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [3 \ 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Observable Canonical Form:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [0 \ 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Diagonal Canonical Form:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [2 \ -1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

14.

EXAMPLE 4.12

Determine the canonical state model of the system, whose transfer function is $T(s) = 2(s+5)/[(s+2)(s+3)(s+4)]$.

SOLUTION

By partial fraction expansion,

$$\begin{aligned} \frac{Y(s)}{U(s)} &= \frac{2(s+5)}{(s+2)(s+3)(s+4)} = \frac{A}{s+2} + \frac{B}{s+3} + \frac{C}{s+4} \\ A &= \left. \frac{2(s+5)}{(s+3)(s+4)} \right|_{s=-2} = \frac{2(-2+5)}{(-2+3)(-2+4)} = \frac{2 \times 3}{1 \times 2} = 3 \\ B &= \left. \frac{2(s+5)}{(s+2)(s+4)} \right|_{s=-3} = \frac{2(-3+5)}{(-3+2)(-3+4)} = \frac{2 \times 2}{-1 \times 1} = -4 \\ C &= \left. \frac{2(s+5)}{(s+2)(s+3)} \right|_{s=-4} = \frac{2(-4+5)}{(-4+2)(-4+3)} = \frac{2 \times 1}{-2 \times (-1)} = 1 \\ \therefore \frac{Y(s)}{U(s)} &= \frac{3}{s+2} - \frac{4}{s+3} + \frac{1}{s+4} \end{aligned} \quad \text{.....(4.12.1)}$$

The state model in matrix form is given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [u] \quad ; \quad y = [3 \quad -4 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

12. What is state space? What are the advantages of state space analysis

- 12 a) Convert the system $\dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u, y = [1 \quad 1]x$ into controllable canonical form by applying similarity transformation. (5)

MODULE 2

1. Obtain the state model of the system whose transfer function is given by $Y(s)/U(s) = 10/[s^3 + 4s^2 + 2s + 1]$
2. Explain any four properties of state transition matrix.
3. Write the solution of non-homogeneous state equation

Q4.29. Write the properties of state transition matrix.

The following are the properties of state transition matrix.

1. $\phi(0) = e^{A \times 0} = I$ (Unit matrix)
2. $\phi(t) = e^{At} = (e^{-At})^{-1} = [\phi(-t)]^{-1}$
3. $\phi(t_1 + t_2) = e^{A(t_1 + t_2)} = e^{At_1} e^{At_2} = \phi(t_1) \phi(t_2) = \phi(t_2) \phi(t_1)$

Write the solution of homogeneous state equations.

The solution of homogeneous state equation is, $X(t) = e^{At} X_0$

where, $X(t)$ = State vector at time, t

e^{At} = State transition matrix

and X_0 = Initial condition vector at $t = 0$.

Write the solution of non-homogeneous state equations.

The solution of non-homogeneous state equation is

$$X(t) = e^{A(t-t_0)} X(t_0) + \int_{t_0}^t e^{A(t-\tau)} B \cdot U(\tau) d\tau$$

where, $X(t)$ = State vector of time, t ; $X(t_0)$ = Initial condition vector at $t = t_0$.

B = Input matrix and $U(t)$ = Input vector

If initial conditions are known at $t = 0$, then put $t_0 = 0$.

4.

Module 2

- 13 a) Obtain the unit step response $y(t)$ of the system

$$\dot{X} = \begin{bmatrix} -1 & 0 \\ -2 & -3 \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} X$$

- b) Show that eigen values of state models are unique.

- 14 a) Determine the state transition matrix for the system with state model:

$$\dot{X} = \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix} X$$

A system is described by $\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -4 & -4 \end{bmatrix} x(t)$.

Determine state transition matrix for the system.

Consider the matrix A . Compute e^{At} by two methods.

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$e^{At}c = \phi(t) = \mathbf{L}^{-1}[(s\mathbf{I} - A)^{-1}]$$

$$s\mathbf{I} - A = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

Let, $\Delta = |s\mathbf{I} - A| = \text{determinant of } (s\mathbf{I} - A)$

$$\therefore \Delta = |s\mathbf{I} - A| = \begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = s(s+3) + 2 = s^2 + 3s + 2 = (s+2)(s+1)$$

$$\phi(s) = [s\mathbf{I} - A]^{-1} = \frac{[\text{Cofactor of } (s\mathbf{I} - A)]^T}{\text{determinant of } (s\mathbf{I} - A)} = \frac{[\text{Cofactor of } (s\mathbf{I} - A)]^T}{\Delta}$$

$$\therefore \phi(s) = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$\phi(s) = \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

By partial fraction expansion, $\phi(s)$ can be written as,

$$\phi(s) = \begin{bmatrix} \frac{A_1}{s+1} + \frac{B_1}{s+2} & \frac{A_2}{s+1} + \frac{B_2}{s+2} \\ \frac{A_3}{s+1} + \frac{B_3}{s+2} & \frac{A_4}{s+1} + \frac{B_4}{s+2} \end{bmatrix}$$

$$\frac{s+3}{(s+1)(s+2)} = \frac{A_1}{s+1} + \frac{B_1}{s+2}$$

$$A_1 = \frac{s+3}{s+2} \Big|_{s=-1} = 2$$

$$B_1 = \frac{s+3}{s+1} \Big|_{s=-2} = -1$$

$$\frac{1}{(s+1)(s+2)} = \frac{A_2}{s+1} + \frac{B_2}{s+2}$$

$$A_2 = \frac{1}{s+2} \Big|_{s=-1} = 1$$

$$B_2 = \frac{1}{s+1} \Big|_{s=-2} = -1$$

$$\frac{-2}{(s+1)(s+2)} = \frac{A_3}{s+1} + \frac{B_3}{s+2}$$

$$A_3 = \frac{-2}{s+2} \Big|_{s=-1} = -2$$

$$B_3 = \frac{-2}{s+1} \Big|_{s=-2} = 2$$

$$\frac{s}{(s+1)(s+2)} = \frac{A_4}{s+1} + \frac{B_4}{s+2}$$

$$A_4 = \frac{s}{s+2} \Big|_{s=-1} = -1$$

$$B_4 = \frac{s}{s+1} \Big|_{s=-2} = 2$$

$$\therefore \phi(s) = \begin{bmatrix} \frac{2}{s+1} - \frac{1}{s+2} & \frac{1}{s+1} - \frac{1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$$

On taking inverse Laplace transform of $\phi(s)$ we get $\phi(t)$, where $\phi(t) = e^{At}$

$$\therefore e^{At} = \phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

A linear time-invariant system is characterized by homogeneous state equation.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Compute the solution of the homogeneous equation, assuming the initial state vector.

$$X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

SOLUTION

$$\text{Here } A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} ; \quad sI - A = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} s-1 & 0 \\ -1 & s-1 \end{bmatrix}$$

$$|sI - A| = \begin{vmatrix} s-1 & 0 \\ -1 & s-1 \end{vmatrix} = (s-1)^2 - 0 = (s-1)^2$$

$$(sI - A)^{-1} = \frac{1}{(s-1)^2} \begin{bmatrix} s-1 & 0 \\ 1 & s-1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s-1} & 0 \\ \frac{1}{(s-1)^2} & \frac{1}{s-1} \end{bmatrix}$$

$$e^{At} = \phi(t) = \mathbf{L}^{-1}[(\phi(s))] = \mathbf{L}^{-1}[(sI - A)^{-1}] = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$$

$$\text{The solution of the state equation is, } X(t) = e^{At} X_0 = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^t \\ te^t \end{bmatrix}$$

Obtain the time response of the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

where $u(t)$ is the unit-step function occurring at $t = 0$, or

$$u(t) = 1(t)$$

For this system,

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The state-transition matrix $\Phi(t) = e^{\mathbf{A}t}$ was obtained in Example 9-5 as

$$\Phi(t) = e^{\mathbf{A}t} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

The response to the unit-step input is then obtained as

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} [1] d\tau$$

or

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

If the initial state is zero, or $\mathbf{x}(0) = \mathbf{0}$, then $\mathbf{x}(t)$ can be simplified to

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

5.3 SIMILARITY TRANSFORMATION

The square matrices **A** and **B** are said to be similar if a nonsingular matrix **P** exists such that

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{B} \quad \text{.....(5.3)}$$

The process of transformation is called similarity transformation and it is a linear transformation. The matrix **P** is called transformation matrix. Also the matrix, **A** can be obtained from **B** by a similarity transformation with a transformation matrix $\mathbf{P}^{-1}$,

$$\text{i.e., } \mathbf{A} = \mathbf{P} \mathbf{B} \mathbf{P}^{-1} \quad \text{.....(5.4)}$$

The similarity transformation can be used for diagonalization of a square matrix. If an $n \times n$ matrix has n linearly independent eigenvectors (i.e., with distinct eigenvalues) then it can be diagonalized by a similarity transformation. If a matrix has multiple eigenvalues then it will not have a complete set of n linearly independent eigenvectors and so it cannot be diagonalized. However such a matrix can be transformed into a Jordan matrix (Jordan canonical form).

The transformation matrix for diagonalization or converting to Jordan form can be obtained from eigenvectors. For a system with n state variables we can find n numbers of eigenvectors $\mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3, \dots, \mathbf{m}_n$. The eigenvectors are column vectors of order $(n \times 1)$. The transformation matrix is obtained by arranging the eigenvectors columnwise as shown in equ(5.5). This transformation matrix is also called *Modal matrix* and denoted by **M**.

$$\text{Modal matrix, } \mathbf{M} = [\mathbf{m}_1 \ \mathbf{m}_2 \ \mathbf{m}_3 \ \dots \ \mathbf{m}_n] \quad \text{.....(5.5)}$$

The similarity transformation will not alter certain properties of the matrix. A property of a matrix is said to be invariant if it is possessed by all similar matrices. The determinant, characteristic equation and trace of a matrix are invariant under a similarity transformation. Since the characteristic equation is invariant the eigenvalues are also invariant under a linear or similarity transformation.

PROOF FOR INVARIANCE OF DETERMINANT

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Let **A** and **B** are similar matrices and **P** be the transformation matrix which transforms **A** to **B** by a similarity transformation, $\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{B}$

$$\therefore \mathbf{B} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} \quad \text{.....(5.6)}$$

On taking determinant of equ(5.6) we get,

$$|\mathbf{B}| = |\mathbf{P}^{-1} \mathbf{A} \mathbf{P}| \quad \text{.....(5.7)}$$

Since the determinant of a product of two or more square matrices is equal to the product of their individual determinants, the equ(5.7) can be written as,

$$\begin{aligned} |\mathbf{B}| &= |\mathbf{P}^{-1}| |\mathbf{A}| |\mathbf{P}| = |\mathbf{A}| |\mathbf{P}^{-1}| |\mathbf{P}| \\ &= |\mathbf{A}| |\mathbf{P}^{-1} \mathbf{P}| = |\mathbf{A}| |\mathbf{I}| & (\because \mathbf{P}^{-1} \mathbf{P} = \mathbf{I}) \\ &= |\mathbf{A}| & (\because |\mathbf{I}| = 1) \end{aligned}$$

From the above analysis it is evident that the determinant of a matrix is invariant under a similarity transformation.

$$|\lambda \mathbf{I} - \mathbf{A}| = |\lambda \mathbf{I} - \mathbf{P} \mathbf{P}^{-1} \mathbf{A} \mathbf{P} \mathbf{P}^{-1}|$$

PROOF FOR INVARIANCE OF CHARACTERISTIC EQUATION AND EIGENVALUES

Let A and B are similar matrices and P be the transformation matrix which transforms A to B by a similarity transformation, $P^{-1}AP = B$.

The characteristic equation of matrix B is given by

$$|\lambda I - B| = 0 \quad \dots(5.8)$$

On substituting $B = P^{-1}AP$ in equ(5.8) we get,

$$\begin{aligned} |\lambda I - B| &= |\lambda I - P^{-1}AP| \\ &= |\lambda P^{-1}P - P^{-1}AP| \quad (\because P^{-1}P = I) \\ &= |P^{-1}(\lambda I - A)P| \quad \dots(5.9) \end{aligned}$$

Since the determinant of a product is the product of the determinant, the equ(5.9) can be written as,

$$\begin{aligned} |\lambda I - B| &= |P^{-1}| |\lambda I - A| |P| = |\lambda I - A| |P^{-1}| |P| \\ &= |\lambda I - A| |P^{-1}P| \\ &= |\lambda I - A| |I| \quad (\because P^{-1}P = I) \\ &= |\lambda I - A| \quad (\because |I| = 1) \end{aligned}$$

From the above analysis it is clear that the characteristic equations of A and B are identical. Since the characteristic equations are identical, the eigenvalues of A and B are identical. Hence the eigenvalues are invariant under a similarity (linearity) transformation.

PROOF FOR INVARIANCE OF TRACE OF A MATRIX

Let A and B are similar matrices and P be the transformation matrix which transforms A to B by a similarity transformation, $P^{-1}AP = B$.

$$\therefore \text{tr } B = \text{tr } P^{-1}AP \quad \dots(5.10)$$

For an $n \times m$ matrix C and $m \times n$ matrix D, regardless of whether $CD = DC$ or $CD \neq DC$, we have,

$$\text{tr}(CD) = \text{tr}(DC) \quad \dots(5.11)$$

Using the property of equ(5.11), the equ(5.10) can be written as,

$$\begin{aligned} \text{tr } B &= \text{tr } AP P^{-1} \\ &= \text{tr } AI = \text{tr } A \quad (\because PP^{-1} = I \text{ and } AI = A) \end{aligned}$$

From the above analysis it is clear that the trace of a matrix is invariant under a similarity transformation.

A linear time invariant system is described by the following state model.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} [u] \quad \text{and} \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Transform this state model into a canonical state model. Also compute the state transition matrix, e^{At} .

SOLUTION

To find eigenvalues

$$\begin{aligned} (\lambda I - A) &= \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \\ &= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} \end{aligned}$$

The characteristic equation is $|\lambda I - A| = 0$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{vmatrix} = \lambda[\lambda(\lambda + 6) + 11] + 1 \times 6 = 0$$

$$\therefore \lambda(\lambda^2 + 6\lambda + 11) + 6 = 0$$

$$\lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$$

$\lambda = -1$ is one of the root of the above equation,

$$\therefore (\lambda + 1)(\lambda^2 + 5\lambda + 6) = 0$$

$$(\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

The eigenvalues are $\lambda = -1, -2, -3$; i.e, $\lambda_1 = -1, \lambda_2 = -2$, and $\lambda_3 = -3$

To find modal matrix

The system matrix is given in the companion form and therefore modal matrix, M is given by vander monde matrix, V.

$$M = V = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

To find M^{-1}

To find canonical form of state model

$$\begin{aligned} \text{Grammian matrix } \Lambda = M^{-1}AM &= \frac{1}{-2} \begin{bmatrix} -6 & -5 & -1 \\ 6 & 8 & 2 \\ -2 & -3 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} \\ &= \frac{1}{-2} \begin{bmatrix} 6 & 5 & 1 \\ -12 & -16 & -4 \\ 6 & 9 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \end{aligned}$$

$$\tilde{B} = M^{-1}B = -\frac{1}{2} \begin{bmatrix} -6 & -5 & -1 \\ 6 & 8 & 2 \\ -2 & -3 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\tilde{C} = CM = [1 \ 0 \ 0] \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} = [1 \ 1 \ 1]$$

The canonical form of state model is,

$$\dot{Z} = \Lambda Z + \tilde{B}U$$

$$Y = \tilde{C}Z + DU \quad (\text{Here } DU \text{ is not defined})$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} [u] \quad ; \quad Y = [1 \ 1 \ 1] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

MODULE 3

Definition of controllability

A system is said to be completely state controllable if it is possible to transfer the system state from any initial state $X(t_0)$ to any other desired state $X(t_f)$ in specified finite time by a control vector $U(t)$.

Kalman's method of testing controllability

Consider a system with state equation, $\dot{X} = AX + BU$. For this system, a composite matrix, Q_c can be formed such that,

$$Q_c = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \quad \dots(5.64)$$

where n is the order of the system (n is also equal to number of state variables)

In this case the system is completely state controllable if the rank of the composite matrix, Q_c is n .

The rank of the matrix is n , if the determinant of $(n \times n)$ composite matrix Q_c is non-zero. i.e, if $|Q_c| \neq 0$, then rank of $Q_c = n$ and the system is completely state controllable.

The advantage in kalman's test is that the calculations are simpler. But the disadvantage in kalman's test is that, we can't find the state variable which is uncontrollable. But in Gilbert's method we can find the uncontrollable state variable which is the state variable corresponding to the row of $\tilde{B}$ which has all zeros.

Definition of observability

A system is said to be completely observable if every state $X(t)$ can be completely identified by measurements of the output $Y(t)$ over a finite time interval.

Kalman's Test for observability

Consider a system with state model, $\dot{X} = AX + BU$; $Y = CX + DU$

For this system, a composite matrix, Q_o can be formed such that,

$$Q_o = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad (A^T)^3 C^T \quad \dots \quad (A^T)^{n-1} C^T] \quad \dots(5.65)$$

where n is the order of the system (n is also equal to number of state variables)

In this case, the system is completely observable if the rank of composite matrix, Q_o is n .

The rank of the matrix is n , if the determinant of $n \times n$ composite matrix Q_o is non-zero. The disadvantage in Kalman's test is that, the non observable state variables cannot be determined.

State the duality between controllability and observability?

Duality property

The concepts of controllability and observability are dual concepts and it is proposed by Kalman as principle of duality.

The principle of duality states that a system is completely state controllable if and only if its dual system is completely observable or vice-versa. [i.e., if the system is observable then its dual is controllable]. Using the principle of duality, the observability of a given system can be checked by testing the state controllability of its dual or vice-versa.

Consider the system S_1 , described by the state model shown below.

$$\dot{X} = AX + BU$$

$$Y = CX$$

Let the dual of system S_1 be denoted as S_2 and the dual system S_2 is described by the following state model.

$$\dot{Z} = A^T Z + C^T V$$

$$N = B^T Z$$

where, Z = State vector of dual system

V = Input vector of dual system ; N = Output vector of dual system

For the system S_1 the composite matrix, Q_{c1} for controllability is given by equ(5.66) and the composite matrix, Q_{o1} for observability is given by equ(5.67).

$$Q_{c1} = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \quad \dots(5.66)$$

$$Q_{o1} = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad \dots \quad (A^T)^{n-1} C^T] \quad \dots(5.67)$$

For the dual system S_2 , the composite matrix, Q_{c2} for controllability is given by equ(5.68) and the composite matrix Q_{o2} for observability is given by equ(5.69) 4

$$Q_{c2} = [C^T \quad A^T C^T \quad (A^T)^2 C^T \quad \dots \quad (A^T)^{n-1} C^T] \quad \dots(5.68)$$

$$Q_{o2} = [A \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \quad \dots(5.69)$$

From equations (5.66) and (5.69) we get $Q_{c1} = Q_{o2}$, hence if the system S_1 is controllable then its dual system S_2 is observable.

From equations (5.67) and (5.68) we get $Q_{o1} = Q_{c2}$, hence if the system S_1 is observable then its dual system S_2 is controllable.

